

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("/ecommerce_customer_data_custom_ratios.csv")
print(df.head())
print(df.info())
print(df.isnull().sum())
df.drop_duplicates(inplace=True)
numeric_cols = df.select_dtypes(include=['int64', 'float64']).columns
for col in numeric_cols:
    df[col] = df[col].fillna(df[col].median())
print(df.describe())
df['Purchase Date'] = pd.to_datetime(df['Purchase Date'])
df['Month'] = df['Purchase Date'].dt.to_period('M').astype(str)
monthly_sales = df.groupby('Month')['Total Purchase Amount'].sum()
plt.figure(figsize=(10,6))
monthly_sales.plot(kind='line', marker='o')
plt.title("Monthly Purchase Trend")
plt.xlabel("Month")
plt.ylabel("Total Purchase Amount")
plt.xticks(rotation=45)
plt.show()
plt.figure(figsize=(10,6))
sns.countplot(x='Churn', data=df)
plt.title("Customer Churn")
plt.xlabel("Churn")
plt.ylabel("Count")
plt.show()
plt.figure(figsize=(10,6))
df['Product Category'].value_counts().plot(kind='bar')
plt.title("Top Product Categories")
plt.xlabel("Product Category")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.show()
plt.figure(figsize=(10,6))
df['Gender'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%'
)
plt.title("Gender Distribution")
plt.ylabel('')
plt.show()
```

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	\
0	46251	2020-09-08 09:38:32	Electronics	12	3	
1	46251	2022-03-05 12:56:35	Home	468	4	
2	46251	2022-05-23 18:18:01	Home	288	2	
3	46251	2020-11-12 13:13:29	Clothing	196	1	
4	13593	2020-11-27 17:55:11	Home	449	1	

	Total Purchase Amount	Payment Method	Customer Age	Returns	\
0	740	Credit Card	37	0.0	
1	2739	PayPal	37	0.0	
2	3196	PayPal	37	0.0	
3	3509	PayPal	37	0.0	
4	3452	Credit Card	49	0.0	

	Customer Name	Age	Gender	Churn
0	Christine Hernandez	37	Male	0
1	Christine Hernandez	37	Male	0
2	Christine Hernandez	37	Male	0
3	Christine Hernandez	37	Male	0
4	James Grant	49	Female	1

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 250000 entries, 0 to 249999

Data columns (total 13 columns):

#	Column	Non-Null Count	Dtype
0	Customer ID	250000 non-null	int64
1	Purchase Date	250000 non-null	object
2	Product Category	250000 non-null	object
3	Product Price	250000 non-null	int64
4	Quantity	250000 non-null	int64
5	Total Purchase Amount	250000 non-null	int64
6	Payment Method	250000 non-null	object
7	Customer Age	250000 non-null	int64
8	Returns	202404 non-null	float64
9	Customer Name	250000 non-null	object
10	Age	250000 non-null	int64
11	Gender	250000 non-null	object
12	Churn	250000 non-null	int64

dtypes: float64(1), int64(7), object(5)

memory usage: 24.8+ MB

None

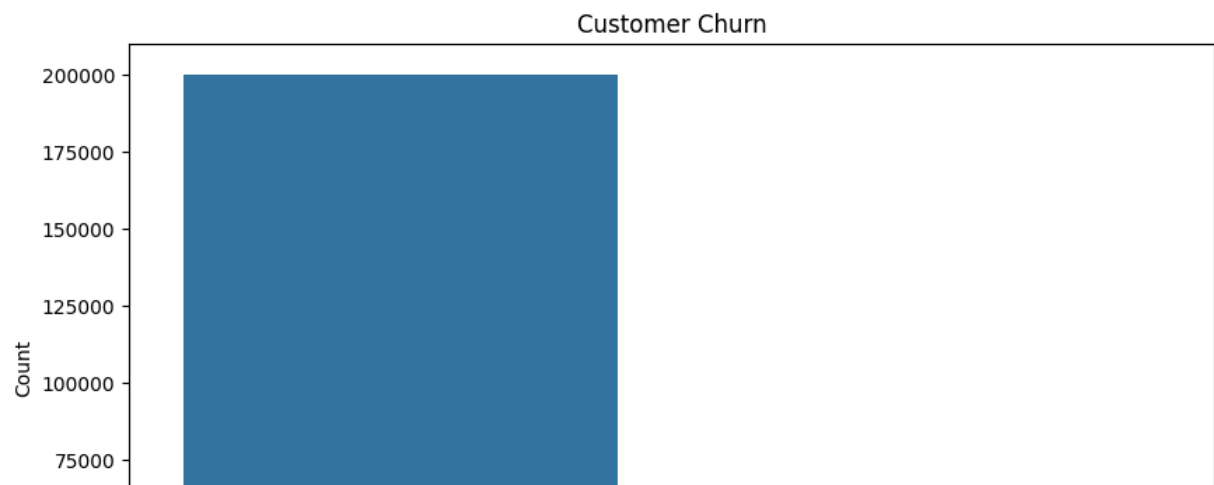
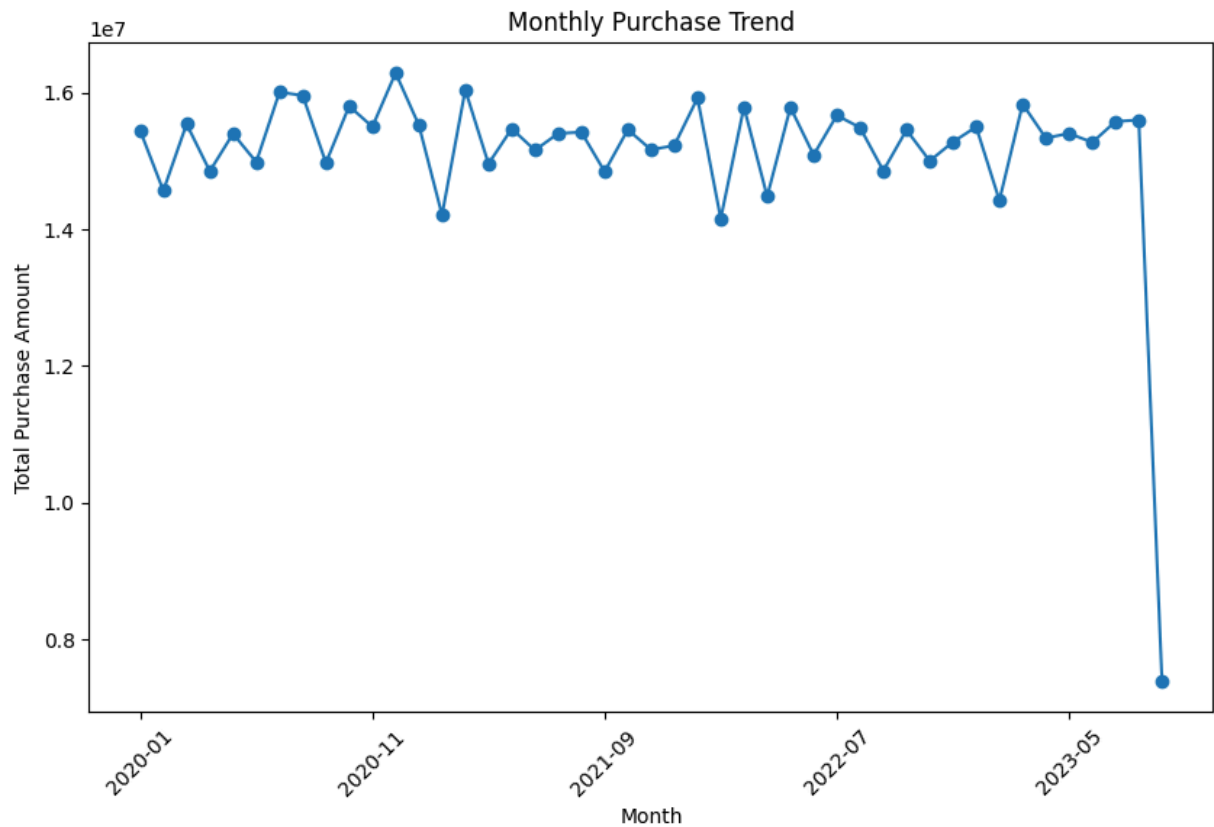
Customer ID	0
Purchase Date	0
Product Category	0
Product Price	0
Quantity	0
Total Purchase Amount	0
Payment Method	0
Customer Age	0
Returns	47596
Customer Name	0
Age	0
Gender	0
Churn	0

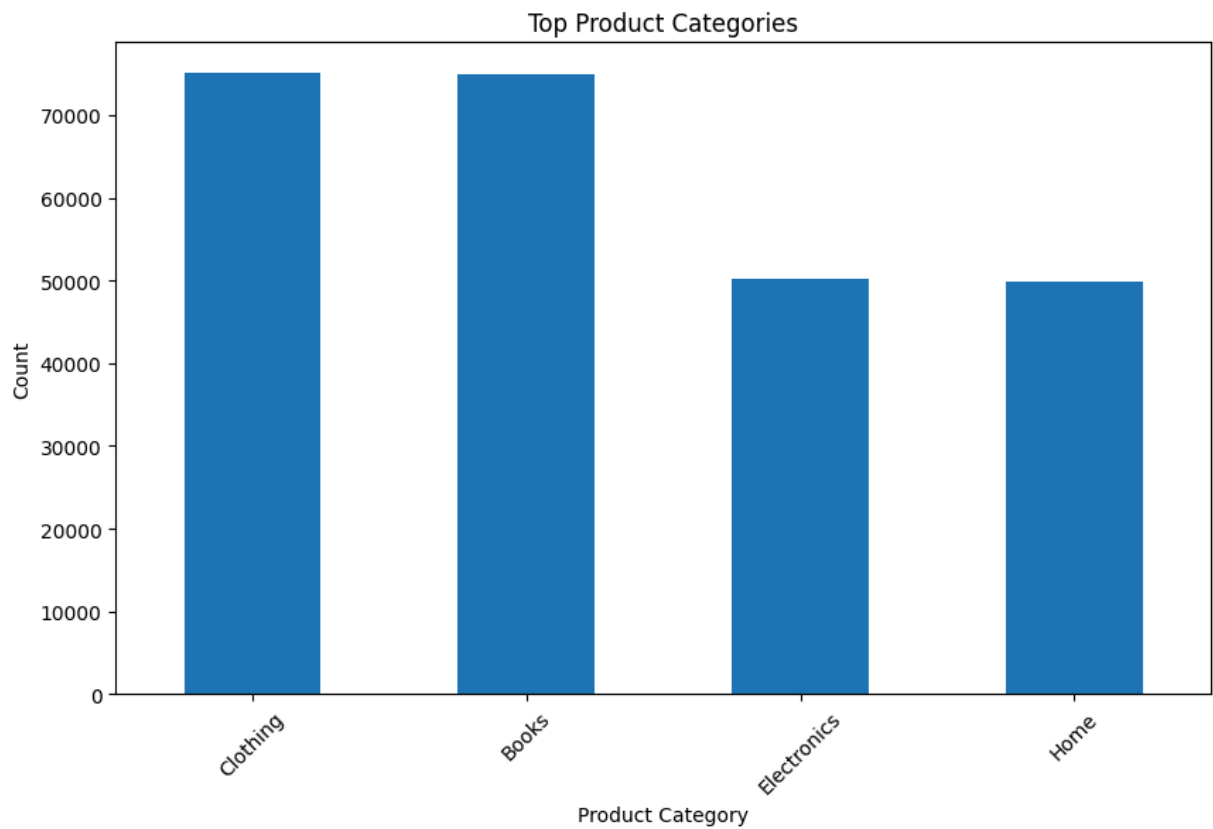
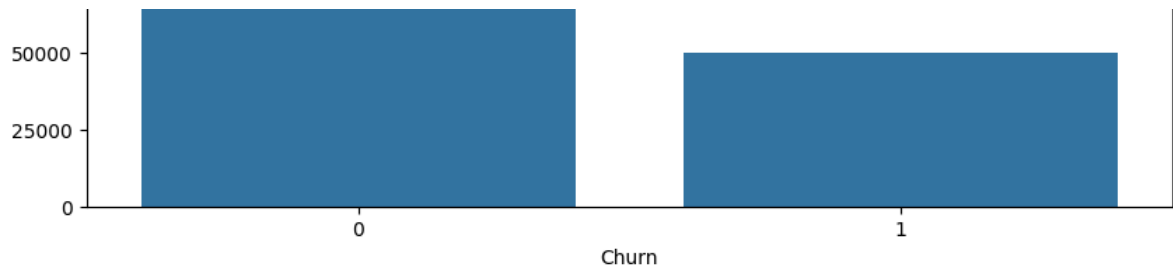
dtype: int64

Customer ID	Product Price	Quantity	Total Purchase Amount	\
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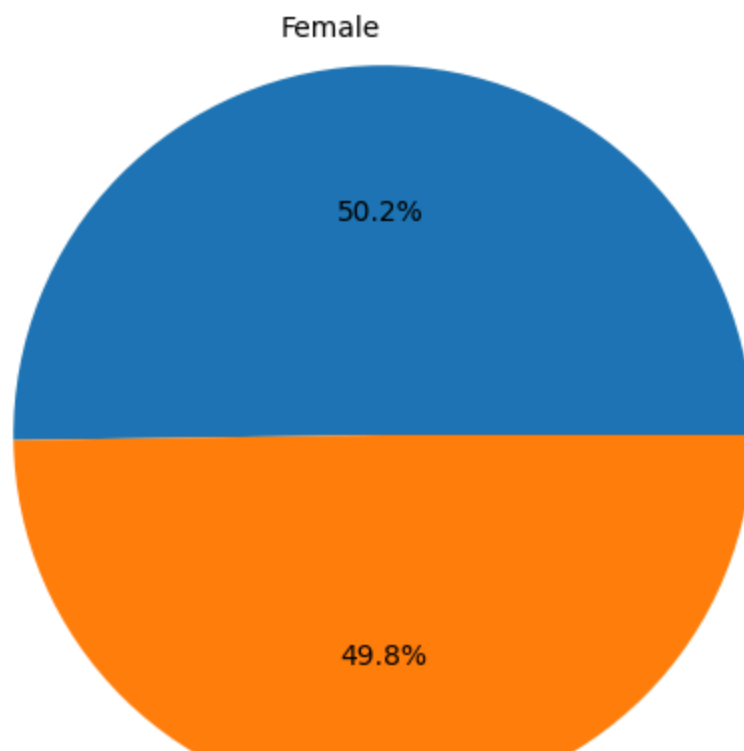
count	250000.000000	250000.000000	250000.000000	250000.000000
mean	25004.03624	254.659512	2.998896	2725.370732
std	14428.27959	141.568577	1.414694	1442.933565
min	1.00000	10.000000	1.000000	100.000000
25%	12497.75000	132.000000	2.000000	1477.000000
50%	25018.00000	255.000000	3.000000	2724.000000
75%	37506.00000	377.000000	4.000000	3974.000000
max	50000.00000	500.000000	5.000000	5350.000000

	Customer Age	Returns	Age	Churn
count	250000.000000	250000.000000	250000.000000	250000.000000
mean	43.940528	0.403076	43.940528	0.199496
std	15.350246	0.490517	15.350246	0.399622
min	18.000000	0.000000	18.000000	0.000000
25%	31.000000	0.000000	31.000000	0.000000
50%	44.000000	0.000000	44.000000	0.000000
75%	57.000000	1.000000	57.000000	0.000000
max	70.000000	1.000000	70.000000	1.000000





Gender Distribution



Customer Behavior Analysis

Male
Objective Analyze customer transactions and behavior to identify purchase patterns and churn risks.

Data Cleaning and Feature Engineering Removed duplicate records Handled missing values using median replacement Converted purchase date into datetime format Created monthly sales feature

Customer Segmentation Customers were analyzed based on: purchase behavior churn trends product category preferences

Purchase Patterns and Retention Trends Electronics and Home categories had high purchases Monthly sales varied across different periods Churn analysis showed retention improvement opportunities

Recommendations for Alfido Tech Focus marketing on popular categories Offer loyalty rewards to repeat customers Use personalized offers to reduce churn Run seasonal promotions during high-sales months Improve customer engagement using targeted campaigns

Conclusion The analysis identified important customer behavior trends and business opportunities that can help improve engagement and retention.